Project Design Phase
Solution Architecture

Date	28 October 2026
Team ID	Individual (Abdul Khadarvali)
Project Name	Employee Raise Issue – Record Producer & Service Portal Integration Guide
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

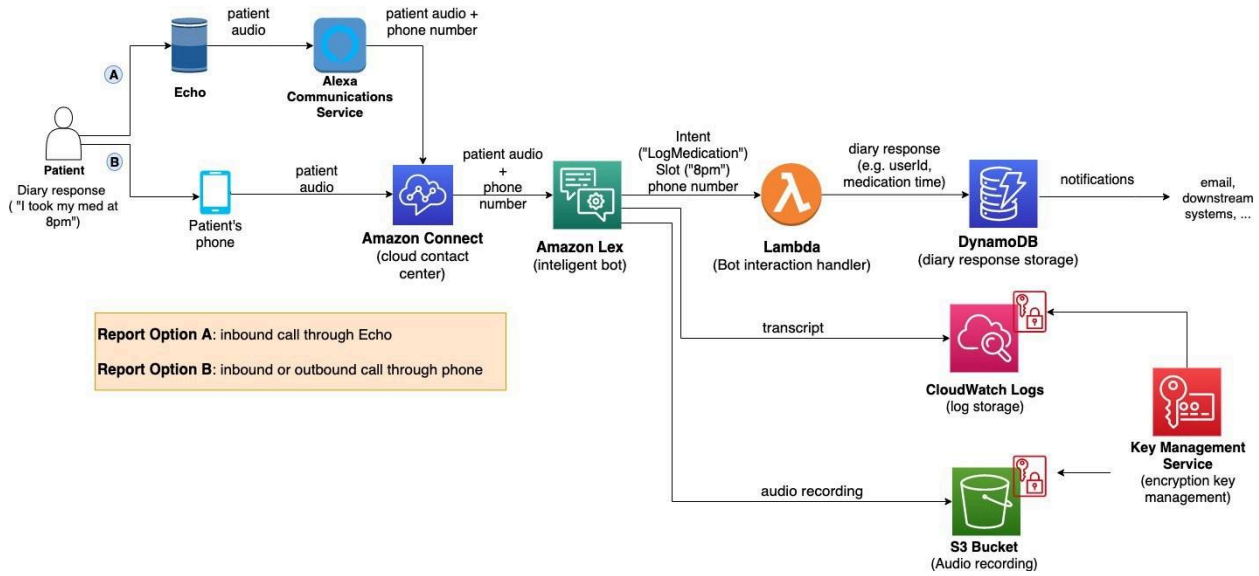


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

Project - Solution Architecture Diagram (Employee Raise Issue System):

End-to-end ServiceNow solution architecture and data flow for the Employee Raise Issue – Record Producer & Service Portal Integration project:

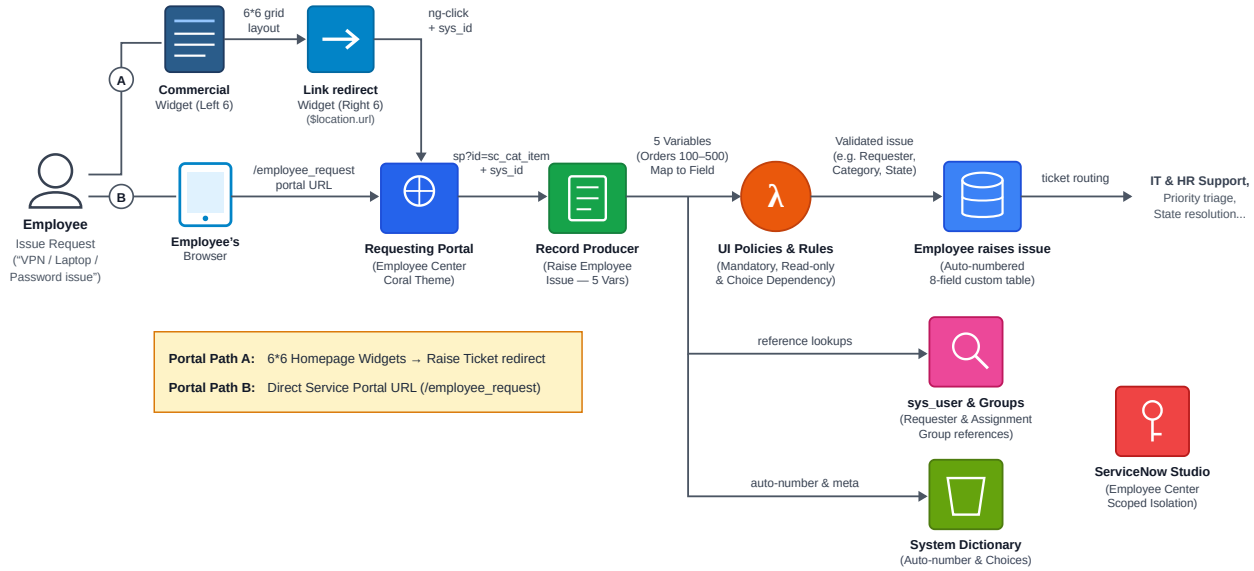


Figure 2: Architecture and data flow of the Employee Raise Issue – Record Producer & Service Portal Integration application

Layered Solution Architecture Specifications:

Architecture Layer	ServiceNow Components	Technical Behavior & Data Flow Specification
1. Presentation Layer (Service Portal)	`Requesting portal` (`/employee_request`), `Employee Center Coral Theme`, `employee_center_portal_page`	Renders the branded portal interface and a 6*6 grid container hosting the left `Commercial widget` (HTML5 + 135-deg CSS gradient) and right `Link redirect` widget (Phases 5 & 6).
2. Client Routing & Intake Layer	`Link redirect` Client Script & `Raise Employee Issue` Record Producer	Clicking `Raise Ticket ->` triggers `$location.url('sp?id=sc_cat_item&sys_id=' + RECORD_PRODUCER_SYS_ID)`, loading 5 mapped variables (`Requester`, `Category`, `Subcategory`, `State`, `Short description` at Orders 100–500) (Phases 4 & 6).
3. Validation & Dependency Layer	2 Table UI Policies & Dictionary Choice Override	Enforces `Short description` as Mandatory when `State is` New, locks `Number` as permanently Read-only, and filters `Subcategory` by `Category` (VPN, `Laptop`, `Server`, `Forgot password`) (Phase 3).
4. Persistence & Reference Layer	`Employee raises issue` Custom Table, `sys_user`, `sys_user_group`	Stores 8 structured fields in the `Employee Center` Scoped Application with system auto-numbering enabled and reference links to requesters and support groups (Phases 1 & 2).